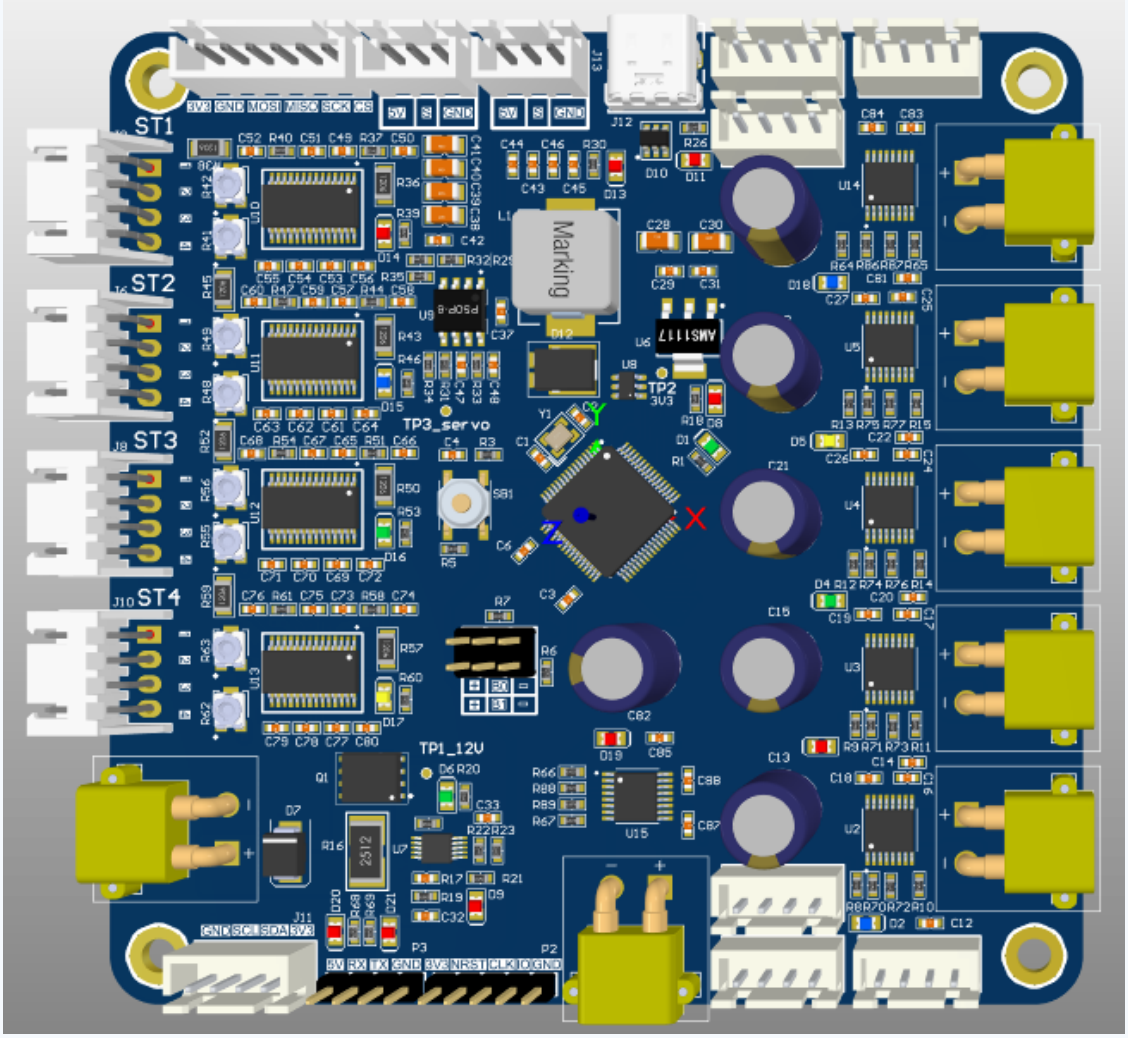
STEERING MOTOR DRIVER

PCB Pinout & Connector Guide



عبدالله بدوي

ABDULLAH BADAWY

ACADEMIC YEAR 2026 / 2027

STM32F401RCT6 | 6 DC MOTORS | 4 STEPPERS

2 SERVOS | 6 ENCODERS | MCU INTERFACES

STEERING MOTOR DRIVER

Motor driver, stepper and servo signal assignment

02

DC Motor Driver Inputs (MCU -> Driver)

MOTOR	SIGNAL	GPIO	PIN #	TIMER ALTERNATE FUNCTION
DC 1	DC1_IN1	PB12	33	GPIO only
DC 1	DC1_IN2	PB13	34	**TIM1_CH1N**
DC 2	DC2_IN1	PB14	35	**TIM1_CH2N**
DC 2	DC2_IN2	PB15	36	**TIM1_CH3N**
DC 3	DC3_IN1	PC6	37	**TIM3_CH1**
DC 3	DC3_IN2	PC7	38	**TIM3_CH2**
DC 4	DC4_IN1	PC8	39	**TIM3_CH3**
DC 4	DC4_IN2	PC9	40	**TIM3_CH4**
DC 5	DC5_IN1	PA8	41	**TIM1_CH1**
DC 5	DC5_IN2	PA9	42	**TIM1_CH2**
DC 6	DC6_IN1	PA5	21	**TIM2_CH1**
DC 6	DC6_IN2	PA6	22	**TIM3_CH1**

Stepper Driver Control Inputs

DRIVER	SIGNAL	GPIO	PIN #	TIMER ALTERNATE FUNCTION
STEP 1	STEP1_IN1	PA0	14	**TIM2_CH1 / TIM5_CH1**
STEP 1	STEP1_IN2	PC0	8	GPIO only
STEP 2	STEP2_IN1	PA1	15	**TIM2_CH2 / TIM5_CH2**
STEP 2	STEP2_IN2	PC1	9	GPIO only
STEP 3	STEP3_IN1	PA2	16	**TIM2_CH3 / TIM5_CH3**
STEP 3	STEP3_IN2	PC2	10	GPIO only
STEP 4	STEP4_IN1	PA3	17	**TIM2_CH4 / TIM5_CH4**
STEP 4	STEP4_IN2	PC3	11	GPIO only

Servo PWM Outputs

SERVO	SIGNAL	GPIO	PIN #	PWM TIMER / CHANNEL
SERVO 1	SERVO_IN1	PB9	62	**TIM4_CH4 / TIM11_CH1**
SERVO 2	SERVO_IN2	PB8	61	**TIM4_CH3 / TIM10_CH1**

NOTE Timer AF means capability, not an already configured firmware function.
One TIMx_CHy output cannot serve two independent hardware PWM signals.

STEERING MOTOR DRIVER

Encoder channels, communication and system signals

03

Encoder Signals: Timer-capable vs GPIO-only

ENCODER	SIGNAL	GPIO	PIN #	TIMER ALTERNATE FUNCTION
ENC 1	ENC1_OUT1	PB10	29	**TIM2_CH3**
ENC 1	ENC1_OUT2	PC14	3	GPIO only
ENC 2	ENC2_OUT1	PA10	43	**TIM1_CH3**
ENC 2	ENC2_OUT2	PC10	51	GPIO only
ENC 3	ENC3_OUT1	PC13	2	GPIO only
ENC 3	ENC3_OUT2	PA15	50	**TIM2_CH1**
ENC 4	ENC4_OUT1	PB1	27	**TIM3_CH4 / TIM1_CH3N**
ENC 4	ENC4_OUT2	PC11	52	GPIO only
ENC 5	ENC5_OUT1	PC5	25	GPIO only
ENC 5	ENC5_OUT2	PB0	26	**TIM3_CH3 / TIM1_CH2N**
ENC 6	ENC6_OUT1	PC4	24	GPIO only
ENC 6	ENC6_OUT2	PA7	23	**TIM3_CH2 / TIM1_CH1N**

Each pair includes a GPIO-only line: use GPIO/EXTI or software decoding, not TIM encoder mode.

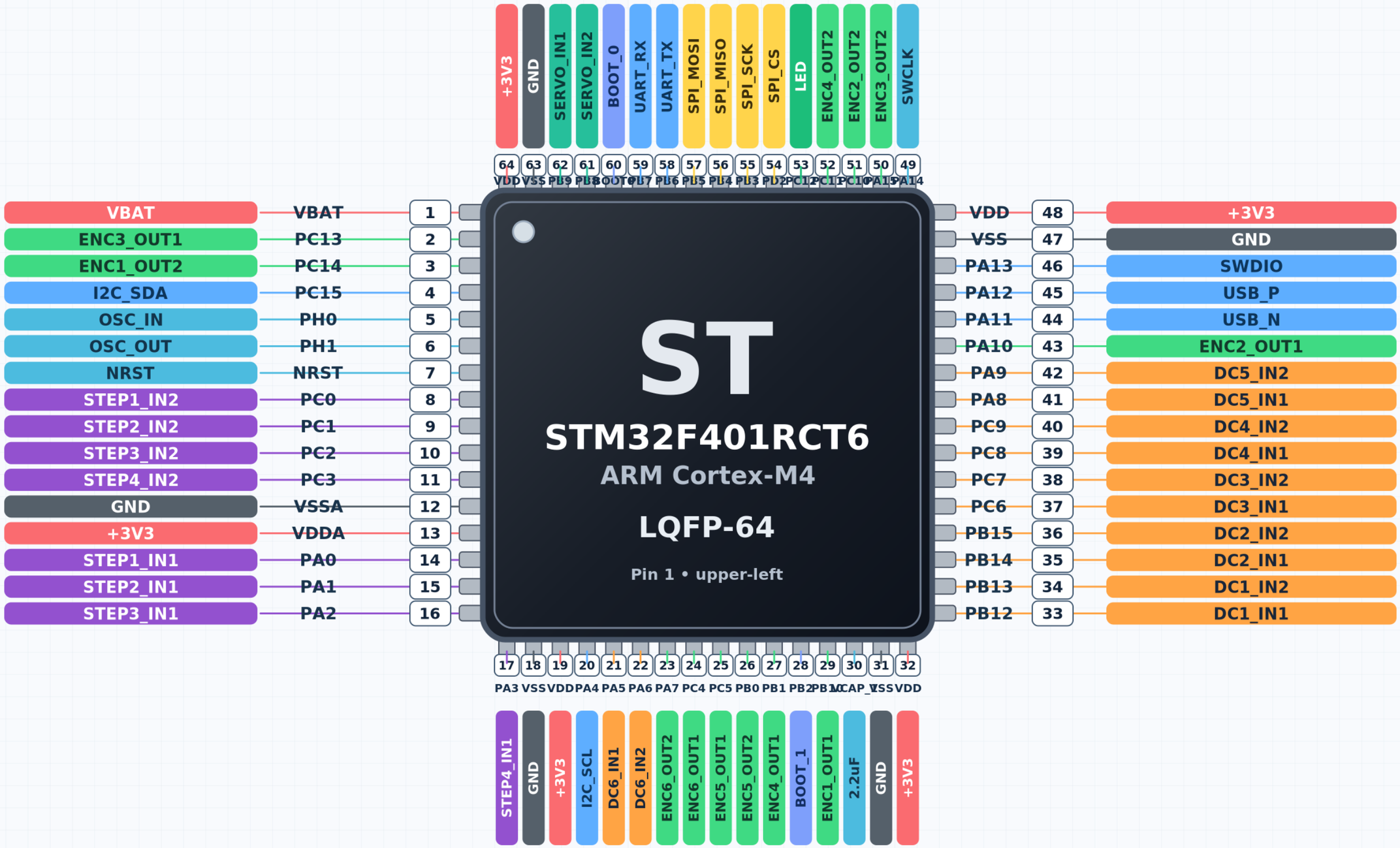
Communication & System Connections

FUNCTION	GPIO	PIN #	INTERFACE / NOTE
UART RX	PB7	59	USART1_RX (AF7), via 1k series R
UART TX	PB6	58	USART1_TX (AF7), via 1k series R
SPI MOSI	PB5	57	SPI1_MOSI (AF5)
SPI MISO	PB4	56	SPI1_MISO (AF5)
SPI SCK	PB3	55	SPI1_SCK (AF5)
SPI CS	PD2	54	GPIO software chip select
I2C SDA	PC15	4	GPIO / software I2C
I2C SCL	PA4	20	GPIO / software I2C
USB D+	PA12	45	USB FS D+ (AF10)
USB D-	PA11	44	USB FS D- (AF10)
SWDIO	PA13	46	Programming / Debug
SWCLK	PA14	49	Programming / Debug
BOOT0	BOOT0	60	Boot configuration pin
BOOT1	PB2	28	Boot option
NRST	NRST	7	MCU reset
USER LED	PC12	53	GPIO output
OSC IN	PH0	5	External HSE oscillator
OSC OUT	PH1	6	External HSE oscillator
VCAP1	VCAP_1	30	2.2 uF to GND (schematic)

Power: VDD/VDDA 3V3, VSS/VSSA GND; VBAT as in schematic.

STM32F401RCT6 — LQFP-64 Pinout

TOP VIEW • PHYSICAL PIN NUMBERS • LATEST SCHEMATIC MAPPING



COLOR KEY

- Step
- DC Motor
- Encoder
- Servo
- SPI
- I2C/UART/USB/SW
- Power
- GND

MAPPING NOTES

I2C: SDA PC15 • SCL PA4 | UART: RX PB7 • TX PB6 (1k)

SPI: MOSI PB5 • MISO PB4 • SCK PB3 • CS PD2

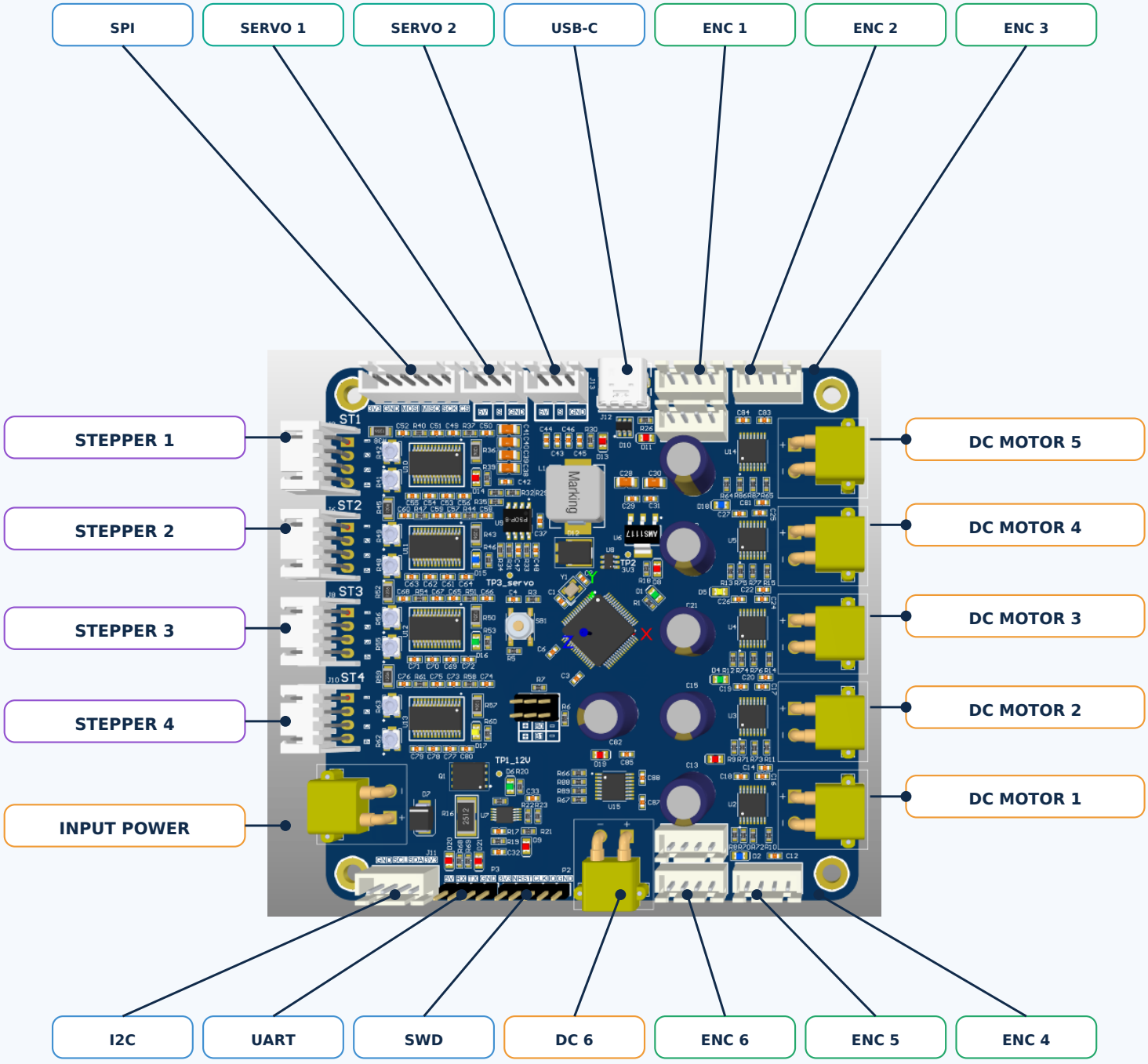
STEERING MOTOR DRIVER

Physical connector location / board top view

05

BOARD CONNECTOR OVERVIEW

Positions are based on the supplied Altium 3D top-view.



Connector labeling reference

Physical connector orientation only. Individual connector contact order, power rails and polarity must be confirmed in the PCB schematic.